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import numpy as np

import matplotlib.pyplot as plt

import pandas as pd

# Load the dataset
dataset = pd.read_csv(r'E:\DataScience\Daily Task and Agendas\22 Sept\Salary_Data.csv')

x = dataset.iloc[:, :-1]

y = dataset.iloc[:, -1]

from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.20, random_state=0)

x_train = x_train.values.reshape(-1, 1)

x_test = x_test.values.reshape(-1, 1)

from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(x_train, y_train)

y_pred = regressor.predict(x_test)

comparison = pd.DataFrame({'Actual': y_test, 'Predicted': y_pred})
print(comparison)

plt.scatter(x_test, y_test, color = 'red')
plt.plot(x_train, regressor.predict(x_train), color = 'blue') # Regression line from training set
plt.title('Salary vs Experience (Test set)')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.show()

plt.scatter(x_train, y_train, color = 'red') # Real salary data (training)
plt.plot(x_train, regressor.predict(x_train), color = 'blue') # Predicted regression line
plt.title('Salary vs Experience (Training set)')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.show()

# ==== best fit line hear ( what next )

coef = print(f"Coefficient: {regressor.coef_}")

intercept = print(f"Intercept: {regressor.intercept_}")

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# future prediction code

exp_12_future_pred = 9312 * 100 + 26780
exp_12_future_pred

bias = regressor.score(x_train, y_train)
print(bias)

variance = regressor.score(x_test, y_test)
print(variance)

# can we implement statistic to this dataset

dataset.mean()

dataset['Salary'].mean()

dataset.median()

dataset['Salary'].median()

dataset['Salary'].mode()

dataset.var()

dataset['Salary'].var()

dataset.std()

# Scipy

from scipy.stats import variation

variation(dataset.values)

variation(dataset['Salary'])

# CO-Relation

dataset.corr()

dataset['Salary'].corr(dataset['YearsExperience'])

# Skewness

dataset.skew()

dataset['Salary'].skew()

dataset.sem()

dataset['Salary'].sem()

import scipy.stats as stats

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dataset.apply(stats.zscore)

stats.zscore(dataset['Salary'])

# anova anlysis of varibale SSE SSR

y_mean=np.mean(y)

SSR=np.sum((y_pred - y_mean) ** 2)

print(SSR)

# SSE

y=y[0:6]
SSE=np.sum((y-y_pred)**2)
print(SSE)

#SST

mean_total=np.mean(dataset.values)
SST= np.sum((dataset.values-mean_total)**2)
print(SST)

# R Squer

r_squer= 1 - (SSR/SST)
r_squer

#Pickle its help to zip all code in one file

import pickle

# Save the trained model to disk
filename = 'linear_regression_model.pkl'

# Open a file in write-binary mode and dump the model
with open(filename, 'wb') as file:
    pickle.dump(regressor, file)

print("Model has been pickled and saved as linear_regression_model.pkl")

import os

print(os.getcwd())

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